

# ECE148: Signal Processing Applications

## Assignment 5: Hilbert Transform

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### 1 Introduction

Consider a real and periodic signal  $f(t)$ , of period  $T$ . The signal has 7 harmonics for  $m = 1, 2, \dots, 7$ .

$$f(t) = \sum_{m=1}^7 a_m \cos\left(\frac{2\pi mt}{T}\right)$$

The 7 coefficients  $a_m$  are formed with my personal 7-digit UCSB perm number,  $[1, 0, 5, 5, 3, 8, 8]$ .

### 2 Method A: Hilbert Transform and Sampling

#### Hilbert Transform of $f(t)$

The Hilbert transform of  $\cos(\omega t)$  is:

$$\mathcal{H}\{\cos(\omega t)\} = \sin(\omega t)$$

Therefore, the Hilbert transform of the periodic function  $f(t)$  is:

$$\hat{f}(t) = \sum_{m=1}^7 a_m \sin\left(\frac{2\pi mt}{T}\right)$$

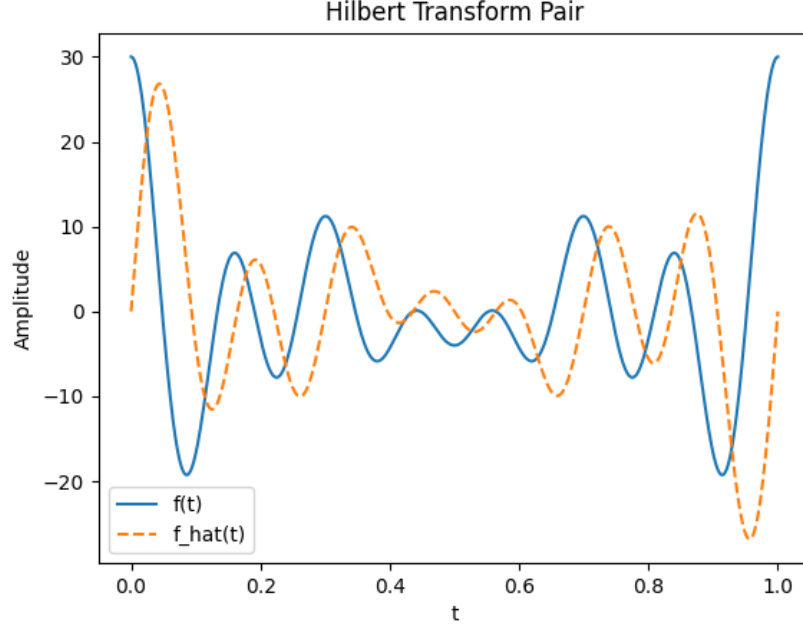


Figure 1: Hilbert transform pair,  $f(t)$  and  $\hat{f}(t)$ , from  $t = 0$  to  $t = T$ .

## Fourier Series Expansion

Using the identities:

$$\cos(\omega t) = \frac{1}{2}(e^{j\omega t} + e^{-j\omega t})$$

$$\sin(\omega t) = \frac{1}{2j}(e^{j\omega t} - e^{-j\omega t})$$

We can formulate both  $f(t)$  and  $\hat{f}(t)$  in the form of complex Fourier series expansion:

$$f(t) = \sum_{m=1}^7 \frac{a_m}{2} e^{j2\pi m t/T} + \sum_{m=1}^7 \frac{a_m}{2} e^{-j2\pi m t/T}$$

$$\hat{f}(t) = \sum_{m=1}^7 \frac{a_m}{2j} e^{j2\pi m t/T} - \sum_{m=1}^7 \frac{a_m}{2j} e^{-j2\pi m t/T}$$

$f(t)$  will have complex coefficients  $C_m = C_{-m} = \frac{a_m}{2j}$ .  $\hat{f}(t)$  will have complex coefficients  $C_m = \frac{a_m}{2j}$  and  $C_{-m} = -\frac{a_m}{2j}$ .

## Orthogonality and Equal Power

$f(t)$  is an even function whereas  $\hat{f}(t)$  is an odd function. An odd function multiplied by an even function is an odd function. The integral of an odd function over one period is zero. Therefore,  $f(t)$  and  $\hat{f}(t)$  are orthogonal.

$$\int_0^T f(t) \hat{f}(t) dt = 0$$

Using Parseval's theorem, we have:

$$\int_{-\infty}^{\infty} |f(t)|^2 dt = \int_{-\infty}^{\infty} f(t) f^*(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\omega) F^*(j\omega) d\omega$$

In the frequency domain,  $\hat{F}(j\omega) = -j \operatorname{sgn}(\omega) F(j\omega)$ . Using Parseval's, we have:

$$\begin{aligned} \int_{-\infty}^{\infty} |\hat{f}(t)|^2 dt &= \int_{-\infty}^{\infty} \hat{f}(t) \hat{f}^*(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{F}(j\omega) \hat{F}^*(j\omega) d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} (-j \operatorname{sgn}(\omega)) F(j\omega) (j \operatorname{sgn}(\omega)) F^*(j\omega) d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\omega) F^*(j\omega) d\omega \end{aligned}$$

Thus, the energies of both signals are the same. In this case, both energies sum to 1504.

## 16-Point Sampling

Given  $\Delta t = \frac{T}{16}$  for  $n = 0, 1, 2, \dots, 15$ , we have:

$$f[n] = f(n\Delta t) = \sum_{m=1}^7 a_m \cos\left(\frac{2\pi m}{16} n\right)$$

$$\hat{f}[n] = \hat{f}(n\Delta t) = \sum_{m=1}^7 a_m \sin\left(\frac{2\pi m}{16} n\right)$$

We check for orthogonality:

$$\sum_{n=0}^{15} f[n] \hat{f}[n] = \sum_{m=1}^7 \sum_{k=1}^7 a_m a_k \sum_{n=0}^{15} \cos\left(\frac{2\pi m}{16} n\right) \sin\left(\frac{2\pi k}{16} n\right)$$

For integer harmonics over one full period,

$$\sum_{n=0}^{15} \cos\left(\frac{2\pi m}{16} n\right) \sin\left(\frac{2\pi k}{16} n\right) = 0$$

Thus, we have  $f[n]$  and  $\hat{f}[n]$  are orthogonal. We solve for the discrete energy of  $f[n]$ :

$$\sum_{n=0}^{15} |f[n]|^2 = \sum_{m,k} a_m a_k \sum_{n=0}^{15} \cos\left(\frac{2\pi m}{16}n\right) \cos\left(\frac{2\pi k}{16}n\right)$$

Using cosine orthogonality, we find that the sum of the two cosines is 8 when  $m = k$ . Therefore, we have the energy:

$$E_f = 8 \sum_{m=1}^7 a_m^2 = 1504$$

Similarly, using sine orthogonality and solving for the discrete energy of  $\hat{f}(t)$  yields:

$$E_{\hat{f}} = 8 \sum_{m=1}^7 a_m^2 = 1504$$

## Analytic Signal and DFT

We form the analytic sequence  $\tilde{f}[n] = f[n] + j\hat{f}[n]$ :

$$\tilde{f}[n] = \sum_{m=1}^7 a_m \cos\left(\frac{2\pi m}{16}n\right) + j \sum_{m=1}^7 a_m \sin\left(\frac{2\pi m}{16}n\right)$$

Using complex exponentials, we have:

$$\tilde{f}[n] = \sum_{m=1}^7 a_m e^{j\frac{2\pi n}{16}m}$$

We solve for the DFT:

$$\begin{aligned} \tilde{F}[k] &= \sum_{m=1}^7 a_m \sum_{n=0}^{15} e^{j\frac{2\pi n}{16}m} e^{-j\frac{2\pi n}{16}k} \\ \tilde{F}[k] &= \sum_{m=1}^7 a_m \sum_{n=0}^{15} e^{-j\frac{2\pi n}{16}(k-m)} \end{aligned}$$

Thus, we have the resultant DFT spectral sequence:

$$[0, 16, 0, 80, 80, 48, 128, 128, 0, 0, 0, 0, 0, 0, 0, 0]$$

## Comparison of $\tilde{F}(k)$ to the 7-point sequence $a_m$

In the continuous-time domain, the original 7-point sequence  $\{a_m\}$  represents the amplitude of the cosine harmonics. The 16-point result  $\tilde{F}[k]$  is directly related to  $a_m$  by the relationship:

$$\tilde{F}[k] = \begin{cases} 16a_k & k = 1, \dots, 7 \\ 0 & \text{otherwise} \end{cases}$$

From the Fourier series expansion, we see that energy is split between  $+m$  and  $-m$ , where each side has  $\frac{a_m}{2}$ . After applying the Hilbert transform and combining the transform pair to form the 16-point complex analytic sequence, we find that the negative frequencies are removed and positive frequencies become full amplitude  $a_m$ . Taking the DFT recovers those amplitudes exactly (scaled by 16).

### 3 Method B: Sampling and Hilbert Transform

#### Sampling $f(t)$

We take 16 uniform samples of  $f(t)$  within one period with sample spacing  $\Delta t = \frac{T}{16}$  to form a 16-point sequence  $f[n] = f(n\Delta t)$ , where  $n = 0, 1, 2, \dots, 15$ .

$$f[n] = f(n\Delta t) = \sum_{m=1}^7 a_m \cos\left(\frac{2\pi m}{16}n\right)$$

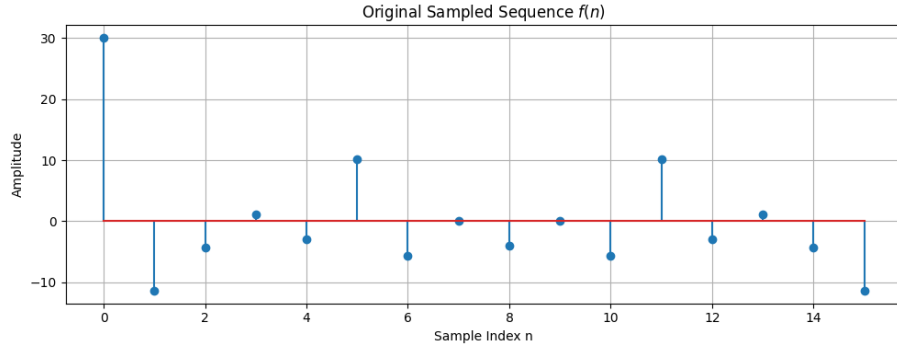


Figure 2: The 16-point sequence  $f[n]$ .

#### Hilbert Transform with 16-Point FFT

The FFT of  $f[n]$  is:

$$F[k] = \sum_{m=1}^7 a_m \sum_{n=0}^{15} \cos\left(\frac{2\pi m}{16}n\right) e^{-j\frac{2\pi n}{16}k}$$

From the FFT, we can use the following identity to derive the FFT of the Hilbert transform.

$$\begin{aligned} \hat{F}[k] &= -j \operatorname{sgn}(k) F[k] \\ \hat{F}[k] &= -j \operatorname{sgn}(k) \sum_{m=1}^7 a_m \sum_{n=0}^{15} \cos\left(\frac{2\pi m}{16}n\right) e^{-j\frac{2\pi n}{16}k} \end{aligned}$$

Taking the inverse DFT yields the discrete-time Hilbert transform  $\hat{f}[n]$ :

$$\hat{f}[n] = \sum_{m=1}^7 a_m \sin\left(\frac{2\pi m}{16}n\right)$$

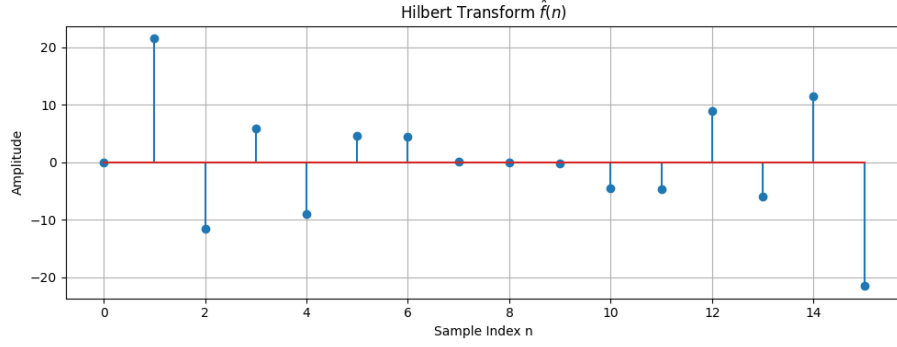


Figure 3: Hilbert Transform of  $f[n]$ .

## Orthogonality and Equal Power

We check the Hilbert transform pair's orthogonality by calculating the inner product:

$$\sum_{n=0}^{15} f[n] \hat{f}[n]$$

The inner product of the pair is 0 as the cosine and sine components are orthogonal to each other. Therefore,  $f[n]$  and  $\hat{f}[n]$  are orthogonal. The discrete energy of the original sequence and the Hilbert transform are given as:

$$E_f = \sum_{n=0}^{15} |f[n]|^2$$

$$E_{\hat{f}} = \sum_{n=0}^{15} |\hat{f}[n]|^2$$

Repeating the same calculations from before, we find that:

$$E_f = E_{\hat{f}} = \sum_{m=1}^7 a_m^2 = 1504$$

Thus, the total energy levels of  $f[n]$  and  $\hat{f}[n]$  are equal to each other.

## Analytic Signal and DFT

Combining the Hilbert transform pair, we form the 16-point complex analytic sequence:

$$\tilde{f}[n] = \sum_{m=1}^7 a_m \cos\left(\frac{2\pi m}{16}n\right) + j \sum_{m=1}^7 a_m \sin\left(\frac{2\pi m}{16}n\right)$$

Taking the 16-point DFT/FFT of the analytic sequence  $\tilde{f}[n]$  results in the following spectral sequence:

$$\tilde{F}[k] = [0, 16, 0, 80, 80, 48, 128, 128, 0, 0, 0, 0, 0, 0, 0, 0]$$

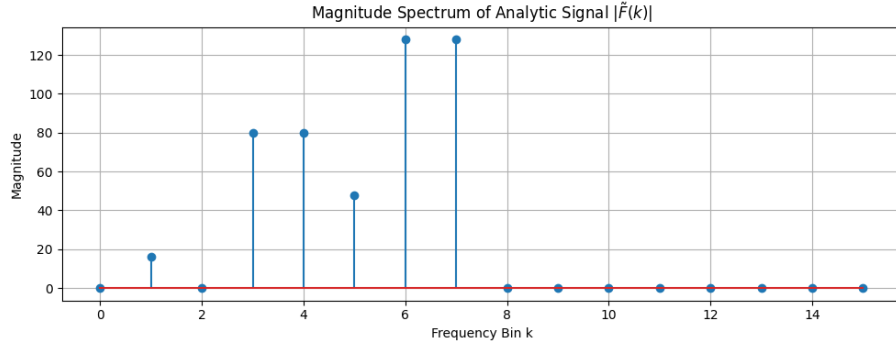


Figure 4: Magnitude spectrum of the analytic signal  $\tilde{F}[k]$ .

## Comparison of $\tilde{F}[k]$ with Part A

In Method A, we applied a Hilbert transform in continuous time and then sampled. In Method B, we sampled first and then applied the Hilbert transform via DFT/FFT. Remarkably, both methods resulted in the same  $\tilde{F}[k]$  where:

$$\tilde{F}[k] = \begin{cases} 16a_k & k = 1, \dots, 7 \\ 0 & \text{otherwise} \end{cases}$$

This occurs because the signal is band-limited and sampled exactly over one period, so no aliasing occurs. The FFT-based Hilbert transform correctly removes negative frequency components and preserves positive frequency components, resulting in the same one-sided spectrum where  $\tilde{F}[k] = 16a_k$  for  $k = 1, \dots, 7$ .